

Expected Rush Yards

Lucca Ferraz, Savan Patel, Ryan Coombs, &
Peyton Stevenson

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The Problem

How well can we predict the number of yards a running back will gain after a handoff? Using player tracking data, we aim to leverage the positions, speeds, and movements of every offensive and defensive player on the field to estimate the expected rushing yards for each play.

The Data

NFL player tracking data from the 2018 season:

- Records position, speed, and direction of every player at the moment of the handoff
- Includes offensive and defensive team information
- Highlights the rusher's position and movement
- Provides full context of each handoff for modeling



Preprocessing

Flip coordinates for
all plays to go
rightward

Identify the rusher in
each play

Label offense and
defensive team for
each player



Adjust player directions and
orientations consistently

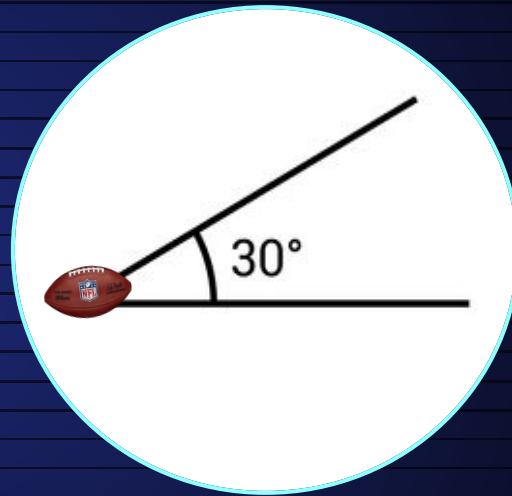
Compute yards from
own goal line

Feature Engineering

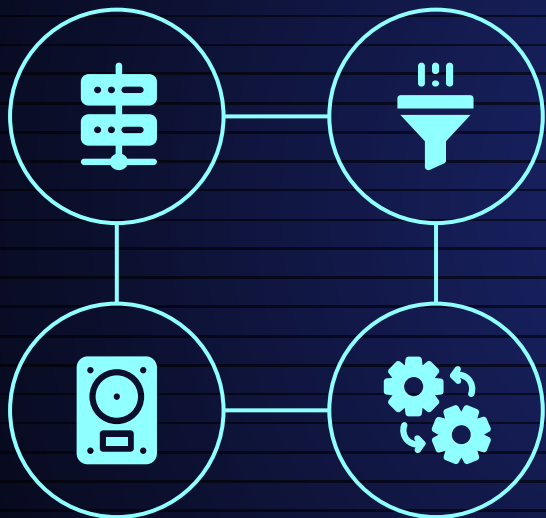
- **Rusher features:** position, speed, acceleration, orientation, direction, distance from line of scrimmage
- **Defender features:** distances to rusher, angles, number ahead in forward cone
- **Blocker features:** number, distances to rusher, line width/depth, average orientation
- **Contextual features:** down, distance, quarter, yard line, score difference, formation, personnel, defenders in the box

Feature Engineering

- **Convex hull features:** area of offensive/defensive players, hull area ratio
- **Cone features:** offensive and defensive hull area in rusher's forward cone, cone area ratio, how many defensive players are in cone, closest cone defender, farthest cone defender



Modeling: XGBoost, but why?



1

Captures complex
non-linear interactions
between players

2

Handles large,
high-dimensional player
tracking features

4

Well-suited for
predicting expected
yards after handoff

3

Robust to missing
values and mixed
feature types



Modeling: Application

- Used engineered features from each play for training
- Target variable: yards gained, converted to 199 classes
- Trained multi-class softmax model for cumulative probabilities
- Handled categorical features with one-hot encoding
- Final model trained on all plays, predictions on test set
- Cross validation for tuning hyperparameters

Takeaways

- Hard to decide what features to use without testing them: some intuitive, some not
- Training and validation take lots of time
- XGBoost performs much better when given more training data
- Tuned hyperparameters increase performance a lot

Conclusion

Submission achieved a CRPS of .01322 with the 15% of the data used.

Results



Summary

We can use a finely tuned XGBoost model to relatively adequately predict expected rush yards in the NFL.

Thank you!



Any questions?

Appendix: Variable Importance

